

7. Directional Derivatives and ∇f

Dr Alan Stoneham
a.stoneham@unsw.edu.au

Directional Derivative

Consider the following topographic map of Nattai National Park. How do we measure the rate of change in elevation **in a particular direction**?

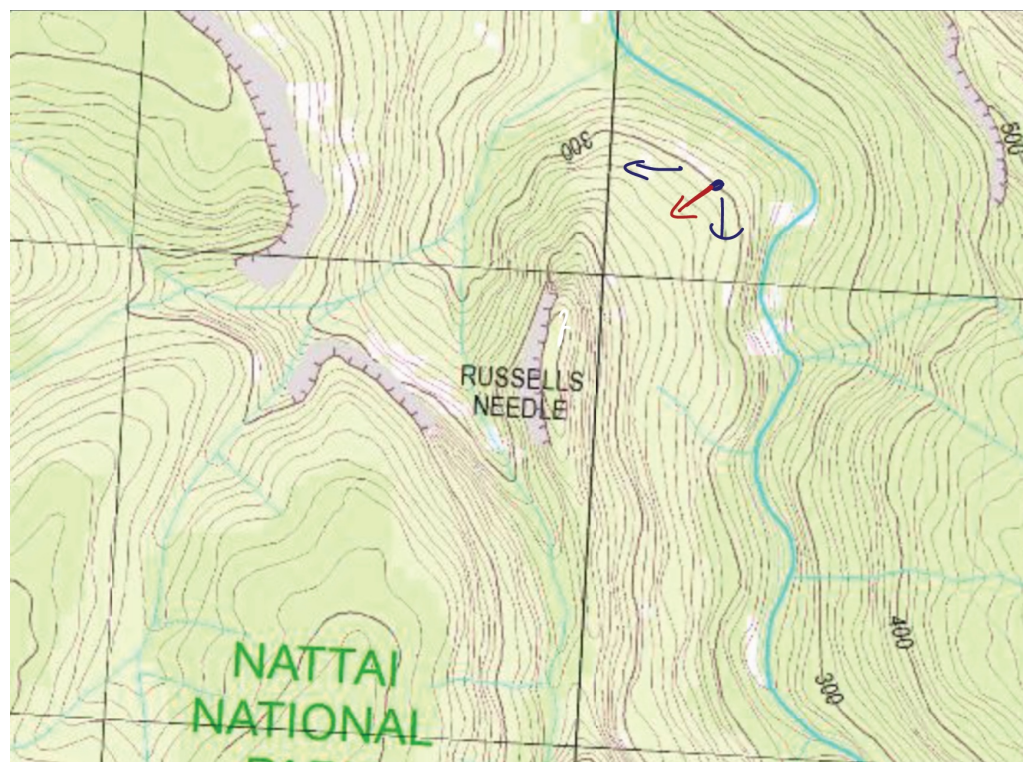
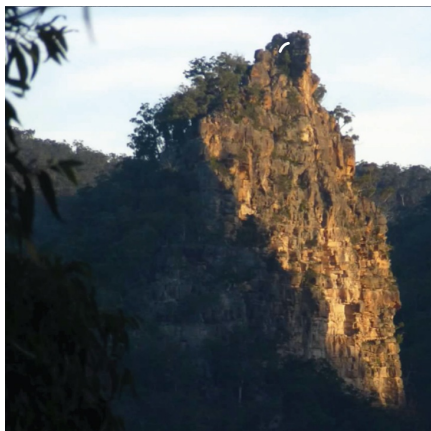


Photo credits: Christian Wilson; fatcanyoners.org

Directional Derivative

Just as $\frac{\partial f}{\partial x_j}$ measures the rate of change in the $\mathbf{e}_j$ direction, we can take inspiration from its definition to measure the rate of change in a given direction.

Definition

Let $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{x}_0 \in \text{Int}(\Omega)$, and $\mathbf{v} \in \mathbb{R}^n$ be non-zero. The **directional derivative of f in the direction $\mathbf{v}$** at $\mathbf{x}_0$, denoted by $D_{\mathbf{v}}f(\mathbf{x}_0)$, is defined by

$$D_{\mathbf{v}}f(\mathbf{x}_0) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}_0 + h\hat{\mathbf{v}}) - f(\mathbf{x}_0)}{h},$$

where $\hat{\mathbf{v}} = \frac{1}{|\mathbf{v}|}\mathbf{v}$.

Some alternative notations include $f'_{\mathbf{v}}(\mathbf{x}_0)$, $\partial_{\mathbf{v}}f(\mathbf{x}_0)$ and $\frac{\partial f}{\partial \mathbf{v}}(\mathbf{x}_0)$.

copying partial derivative definition
but $\hat{\mathbf{v}}$ instead of $\mathbf{e}_j$

Directional Derivative

Remarks:

- $D_{\underline{u}} f(\underline{x}) = \frac{\partial f}{\partial x_j}(\underline{x}_0)$ for all $1 \leq j \leq n$.
- $\underline{x}_0 \in \text{Int}(\Omega)$ is not necessary. we only need that there is $\delta > 0$ st. $|h| < \delta$, then $\underline{x}_0 + h \underline{\hat{u}} \in \Omega$.
- when such a $\delta > 0$ exist, can define $\gamma = (-\delta, \delta)$ by $g(t) = f(\underline{x}_0 + t \underline{\hat{u}})$, so that $\gamma \in \mathbb{R}$
$$g'(0) = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{f(\underline{x}_0 + h \underline{\hat{u}}) - f(\underline{x}_0)}{h}$$
$$= D_{\underline{u}} f(\underline{x}_0)$$

Computing the Directional Derivative

Suppose that f is differentiable. $f: \mathbb{R}^n \rightarrow \mathbb{R}$

Let's try calculating $D_{\underline{v}} f(\underline{x}_0)$. Parameterise the line segment through $\underline{x}_0$ and parallel to $\underline{v}$ by $\underline{c}(t) = \underline{x}_0 + t\underline{v}$. Then

$$D_{\underline{v}} f(\underline{x}_0) = \lim_{h \rightarrow 0} \frac{f(\underline{x}_0 + h\underline{v}) - f(\underline{x}_0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(\underline{c}(h)) - f(\underline{c}(0))}{h}$$

$$= D(f \circ \underline{c})(0)$$

$$= Df(\underline{c}(0)) \underline{c}'(0) = Df(\underline{x}_0) \cdot \underline{\hat{v}}$$

$$= \begin{pmatrix} \frac{\partial f}{\partial x_1}(\underline{x}_0) \\ \frac{\partial f}{\partial x_2}(\underline{x}_0) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\underline{x}_0) \end{pmatrix} \cdot \underline{\hat{v}}$$

Computing the Directional Derivative

$$\nabla f = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} f$$

Definition *Codomain = $\mathbb{R}$*

Let $f : \Omega \subseteq \mathbb{R}^n$ and $\mathbf{x}_0 \in \text{Int}(\Omega)$. The **gradient of f at $\mathbf{x}_0$** , denoted by $\nabla f(\mathbf{x}_0)$ or $\text{grad}(f)(\mathbf{x}_0)$, is the vector

$$\nabla f(\mathbf{x}_0) = \left(\frac{\partial f}{\partial x_1}(\mathbf{x}_0), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}_0) \right). \quad = J_f(\mathbf{x}_0)$$

Theorem

Let $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $\mathbf{x}_0 \in \text{Int}(\Omega)$. If $\mathbf{v} \in \mathbb{R}^n$ is non-zero, then $D_{\mathbf{v}}f(\mathbf{x}_0)$ exists and

$$D_{\mathbf{v}}f(\mathbf{x}_0) = \nabla f(\mathbf{x}_0) \cdot \hat{\mathbf{v}}.$$

Computing the Directional Derivative

Exercise. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by $f(x, y, z) = x^2 e^{-yz^4}$. Find the rate of change of f in the direction of $\mathbf{v} = (1, 1, 1)$ at the point $(1, 1, -1)$.

Note that f is differentiable. setting $\underline{x} = (1, 1, 1)$ and $\underline{x}_0 = (1, 1, -1)$, then $D_{\underline{x}} f(\underline{x}_0) = \nabla f(\underline{x}_0) \cdot \underline{v}$

$$\frac{\partial f}{\partial x} = 2x e^{-yz^4} \quad \left| \quad \frac{\partial f}{\partial y} = -x^2 z^4 e^{-yz^4} \right.$$

$$\frac{\partial f}{\partial z} = -4x^2 y z^3 e^{-yz^4}$$

$$\nabla f(1, 1, -1) = \left(\frac{2}{e}, -\frac{1}{e}, \frac{4}{e} \right)$$

so

$$\begin{aligned} D_{\underline{x}} f(\underline{x}_0) &= \frac{1}{e} \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \\ &= \frac{5}{e\sqrt{3}} \end{aligned}$$

$\uparrow$
 $\underline{v}$

Computing the Directional Derivative $\frac{\partial}{\partial x} \frac{1}{3} (x^3 + y^3)^{-\frac{2}{3}} (3x^2)$

Exercise. Let $\mathbf{v} = (1, 2)$. Find the partial derivatives of $f(x, y) = (x^3 + y^3)^{1/3}$ at $(0, 0)$, calculate $\nabla f(0, 0) \cdot \hat{\mathbf{v}}$, and then find the directional derivative of f at $(0, 0)$ in the direction of $(1, 2)$ from the definition.

↑
undefined
at 0?

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt[3]{h^3}}{h} = 1$$

similarly

$$\frac{\partial f}{\partial y}(0, 0) = 1$$

$$\text{so } \nabla f(0, 0) \cdot \hat{\mathbf{v}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{3}{\sqrt{5}}.$$

Computing the Directional Derivative

Ex. cont. Let $\mathbf{v} = (1, 2)$. Find the partial derivatives of $f(x, y) = (x^3 + y^3)^{1/3}$ at $(0, 0)$, calculate $\nabla f(0, 0) \cdot \hat{\mathbf{v}}$, and then find the directional derivative of f at $(0, 0)$ in the direction of $(1, 2)$ from the definition.

$$\begin{aligned} D_{\hat{\mathbf{v}}} f(0, 0) &= \frac{\lim_{h \rightarrow 0} f(h\hat{\mathbf{v}}) - f(\underline{0})}{h}, & \underline{x}_0 &= (0, 0) = \underline{0} \\ &= \lim_{h \rightarrow 0} \frac{f\left(\frac{h}{\sqrt{5}}, \frac{h}{\sqrt{5}}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left(\frac{h^3}{\sqrt{5}^3} + \frac{2^3 h^3}{\sqrt{5}^3}\right)^{1/3}}{h} & \nabla f(\underline{x}_0) \cdot \hat{\mathbf{v}} \\ &= \lim_{h \rightarrow 0} \frac{\frac{h}{\sqrt{5}} 9^{1/3}}{h} = \frac{9^{1/3}}{\sqrt{5}} \neq \frac{3}{\sqrt{5}} & // \end{aligned}$$

(This implies f is NOT differentiable at $\underline{0}$)

Properties of the Directional Derivative

Suppose that $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\mathbf{x}_0 \in \Omega$ and $\mathbf{v} \in \mathbb{R}^n$ is non-zero, then

if $D_{\mathbf{v}} f(\mathbf{x}_0)$ exist, then

$$D_{\mathbf{v}} (f(\mathbf{x}_0)) = \nabla f(\mathbf{x}_0) \cdot \hat{\mathbf{v}} = |\nabla f(\mathbf{x}_0)| \cos \theta$$

where θ is the angle between $\hat{\mathbf{v}}$ and $\nabla f(\mathbf{x}_0)$.

- This is maximised when $\theta = 0$, or when $\hat{\mathbf{v}} \parallel \nabla f(\mathbf{x}_0)$. i.e. $\nabla f(\mathbf{x}_0)$ points in the direction of greatest change, or "steepest ascent".
- Similarly, $-\nabla f(\mathbf{x}_0)$ points in the direction of greatest descent.
- if $\theta = \frac{\pi}{2}$, so $\hat{\mathbf{v}} \perp \nabla f(\mathbf{x}_0)$, then $D_{\mathbf{v}} f(\mathbf{x}_0) = 0$

Back to Tangent Planes

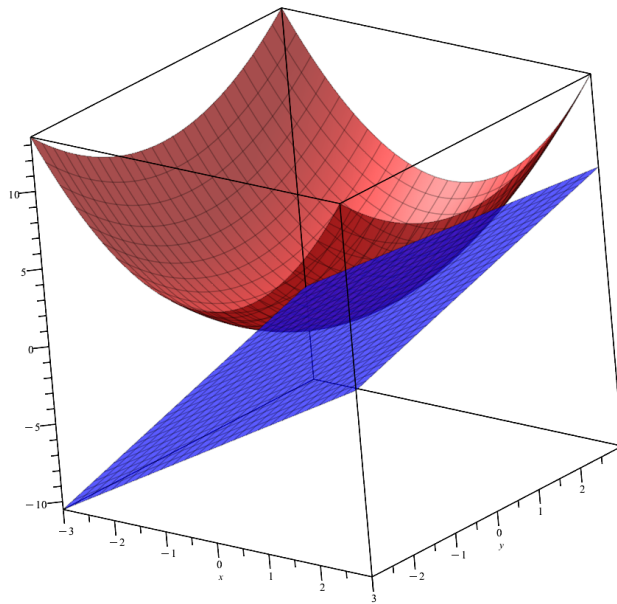
Recall that for a function $f : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $\mathbf{x}_0$, the best affine approximation $T_{\mathbf{x}_0}f : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$\begin{aligned} T_{\mathbf{x}_0}f(\mathbf{x}) &= J_f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + f(\mathbf{x}_0) \\ &= \nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0) + f(\mathbf{x}_0) \end{aligned}$$

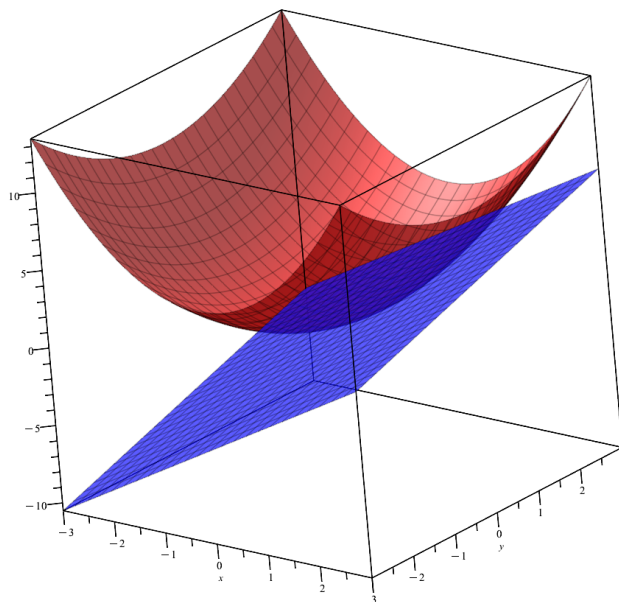
and describes the tangent plane to graph of $z = f(\mathbf{x})$ at $\mathbf{x}_0$.

Recall that the best affine approximation of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + \frac{1}{2}y^2$ at $(1, 1)$ is

$$T_{(1,1)}f(x, y) = 2(x - 1) + (y - 1) + \frac{3}{2}.$$



Back to Tangent Planes



Thus the **tangent plane** to the **graph of $z = f(x, y)$** can be written as

$$TP_f(\mathbf{x}_0) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 3/2 \end{pmatrix} \right) = 0 \right\}.$$

Back to Tangent Planes

$$TP_f(\mathbf{x}_0) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 3/2 \end{pmatrix} \right) = 0 \right\}.$$

The **normal** to the surface at $\mathbf{x}_0 = (1, 1, 3/2)$ is given by

$$\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x}(1, 1) \\ \frac{\partial f}{\partial y}(1, 1) \\ -1 \end{pmatrix} = \nabla \phi(\underline{x}_0)$$

where $\phi(x, y, z) = f(x, y) - z$ and

$\phi(x, y, z) = 0$ defines the surface implicitly.

Back to Tangent Planes

More generally, consider an implicitly defined surface $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^n : \phi(\mathbf{x}) = 0\}$, where $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable, and a point $\mathbf{x}_0 \in \mathcal{S}$.

if $\underline{c} : I \rightarrow \mathcal{S}$ is a differentiable curve
along $\mathcal{S}$ such that $\underline{c}(0) = \underline{x}_0$, $0 \in \text{Int}(I)$,
then

$$\phi(\underline{c}(t)) = 0 \quad \forall t \in I$$

Chain
Rule

$$\Rightarrow D(\phi \circ \underline{c})(t) = 0 \quad \forall t \in I$$

$$\Rightarrow \nabla \phi(\underline{c}(t)) \cdot \underline{c}'(t) = 0 \quad \forall t \in I$$

$$\Rightarrow \nabla \phi(\underline{x}_0) \cdot \underline{c}'(0) = 0.$$



so $\nabla \phi(\underline{x}_0) \perp \underline{c}'(0)$ for all such curves $\underline{c}$.

As $\underline{c}'(0)$ is tangent to $\mathcal{S}$, then $\nabla \phi(\underline{x}_0)$ is normal
to $\mathcal{S}$ at $\underline{x}_0$.

Tangent Plane to an Implicitly Defined Surface

Theorem

Suppose that a surface $\mathcal{S} \subseteq \mathbb{R}^n$ is defined implicitly by $\phi(\mathbf{x}) = 0$ where $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous. If ϕ is differentiable at $\mathbf{x}_0 \in \mathcal{S}$, then the tangent plane to $\mathcal{S}$ at $\mathbf{x}_0$ is

$$T = T(\phi, \mathbf{x}_0) := \{\mathbf{x} \in \mathbb{R}^n : \nabla \phi(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0) = 0\}.$$

Exercise. Find a Cartesian equation for the tangent plane to the surface defined by $x^2 - xyz + z^3 = 2$ at the point $(1, 2, -1)$.

$$\text{Let } \phi(x, y, z) = x^2 - xyz + z^3, \quad \underline{x}_0 = (1, 2, -1)$$
$$\nabla \phi = \begin{pmatrix} 2x - yz \\ -xz \\ 3x^2 - xy \end{pmatrix}, \quad \text{so } \nabla \phi(\underline{x}_0) = \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}.$$

$$\begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right) = 0$$

$$\Leftrightarrow 4x + y + z = 5.$$

Tangent Lines

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ which is differentiable at $\underline{a}$, then its tangent plane is given by the Cartesian equation

$$z = f(\underline{a}) + \frac{\partial f}{\partial x_1}(\underline{a})(x_1 - a_1) + \dots + \frac{\partial f}{\partial x_n}(\underline{a})(x_n - a_n)$$

In parametric vector form, we could write

$$= f(\underline{a}) + \nabla f(\underline{a}) \cdot (\underline{x} - \underline{a})$$

$$\underline{x} \in \mathbb{R}^n = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \\ f(\underline{a}) \end{pmatrix} + \lambda_1 \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{\partial f}{\partial x_1}(\underline{a}) \end{pmatrix} + \dots + \lambda_n \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{\partial f}{\partial x_n}(\underline{a}) \end{pmatrix}$$

Setting $\lambda_i = 0$ except for when $i = j$ gives the parametric equation of the tangent line along the x_j -axis:

$$\underline{x} = \underbrace{(\underline{a}, f(\underline{a}))}_{(a_1, a_2, \dots, a_n, f(\underline{a}))} + \lambda_j (a_j + f_{x_j}(\underline{a}) \underline{e}_{n+1})$$

Tangent Lines

The direction vector of the tangent line being $\mathbf{e}_j + f_{x_j}(\mathbf{a})\mathbf{e}_{n+1}$ tells us that for every unit travelled in the $\mathbf{e}_j$ direction, $f_{x_j}(\mathbf{a})$ units are travelled in the z (or x_{n+1}) direction.

What about a tangent line in a more general direction $\mathbf{v}$?

we could get that tangent line should have equation

$$\underline{x} = (\underline{a}, f(\underline{a})) + \lambda (\underline{\hat{v}}, 0) + D_{\underline{v}}f(\underline{a})\underline{e}_{n+1},$$

$\lambda \in \mathbb{R}$

Recall that the graph of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the set

$$\{(\underline{x}, f(\underline{x})) : \underline{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^{n+1}$$

We want to consider the graph of f but only along the $\mathbf{v}$ -axis by intersecting the graph of f with the plane defined by

$$\{(\underline{a}, f(\underline{a})) + \alpha (\underline{\hat{v}}, 0) + \beta \underline{e}_{n+1} : \alpha, \beta \in \mathbb{R}\}$$

Tangent Lines

$$(\underline{a} + \alpha \hat{v}, f(\underline{a}) + \beta)$$

Doing so gives

$$\begin{aligned} & \{(\underline{x}, f(\underline{x})) : \underline{x} \in \mathbb{R}^n\} \cap \{(\underline{a}, f(\underline{a})) + \alpha(\hat{v}, 0) + \beta \underline{e}_{n+1} : \alpha, \beta \in \mathbb{R}\} \\ &= \{(\underline{a} + \alpha \hat{v}, f(\underline{a} + \alpha \hat{v})) : \alpha \in \mathbb{R}\} + \beta \underline{e}_{n+1} : \alpha, \beta \in \mathbb{R} \\ &= \{(\underline{a} + t \hat{v}, g(t)) : t \in \mathbb{R}\}, \text{ where } g(t) = f(\underline{a} + t \hat{v}) \end{aligned}$$

This intersection is a curve that can be parameterised by $c : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by

$$\underline{c}(t) = \underline{a} + (t \hat{v}, g(t))$$

Differentiating c gives

$$c'(t) = (\hat{v}, g'(t))$$

$$c'(0) = (\hat{v}, D_{\hat{v}} f(\underline{a})) \quad g'(0) = D_{\hat{v}} f(\underline{a})$$

That is, $(\hat{v}, D_{\hat{v}} f(\underline{a}))$ is tangent to the surface at $(\underline{a}, f(\underline{a}))$.

Tangent Line Example

Example. Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = xy^2 - x + x^4y^3$, which is differentiable everywhere, with

$$\nabla f = \begin{pmatrix} y^2 - 1 + 4x^3y^3 \\ 2xy + 3y^2x^4 \end{pmatrix}$$

Then the directional derivative at $(0, 0)$ in the direction of $\mathbf{v} = (1, 1)$ is

$$D_{\mathbf{v}}f(0, 0) = \nabla f(0, 0) \cdot \hat{\mathbf{v}} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -\frac{1}{\sqrt{2}}$$

Now define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(t) = f(\mathbf{0} + t\hat{\mathbf{v}}) = f\left(\frac{t}{\sqrt{2}}, \frac{t}{\sqrt{2}}\right) = \frac{t^3}{2\sqrt{2}} - \frac{t}{\sqrt{2}} + \frac{t^7}{8\sqrt{2}}$

We can calculate $g'(0)$:

$$g'(0) = -\frac{1}{\sqrt{2}}$$

and confirm that $g'(0) = D_{\mathbf{v}}f(0, 0)$.

Tangent Line Example

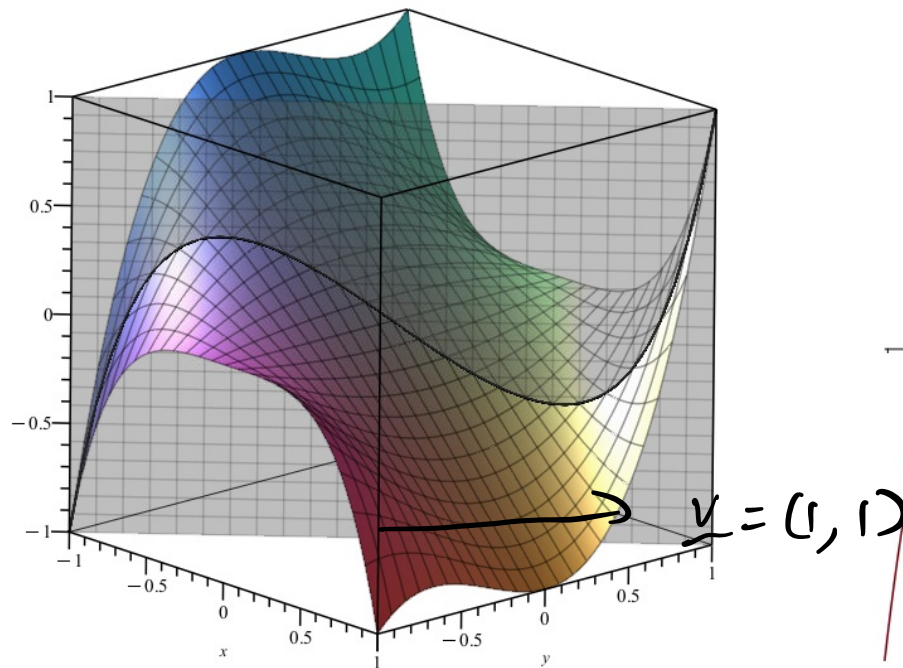


Figure: Plot of $f(x, y)$ for $-1 \leq x, y \leq 1$ with the plane $x = y$.

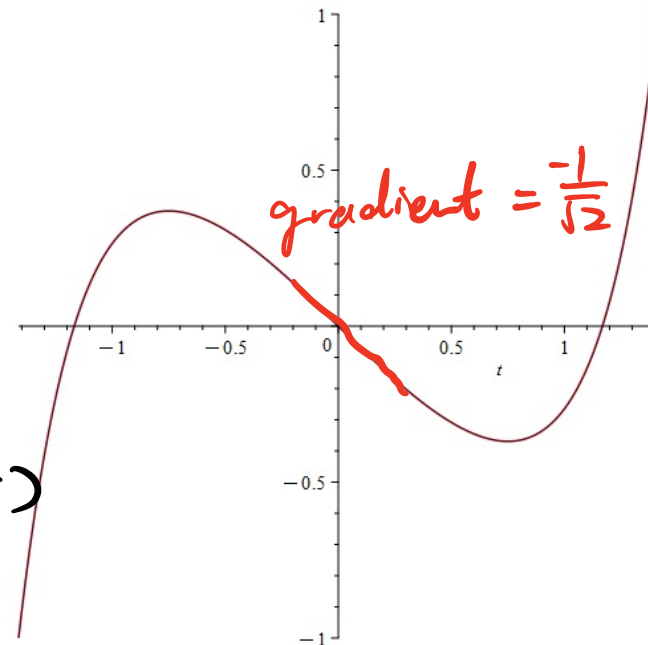


Figure: Plot of $g(t)$ for $-\sqrt{2} \leq t \leq \sqrt{2}$.

The graph of g can be considered as some sort of projection of the graph of f intersected with the plane.

Tangent Lines

$$= (\hat{v}_1, \hat{v}_2, D_{\underline{v}}(0,0)) \in \mathbb{R}^3$$

Thus $(\hat{v}_1, D_{\underline{v}}(0,0)) = \frac{1}{\sqrt{2}}(1,1)$

and the tangent line is

$$\underline{x} = \frac{\lambda}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$$

or

$$\underline{x} = \mu \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mu \in \mathbb{R}$$

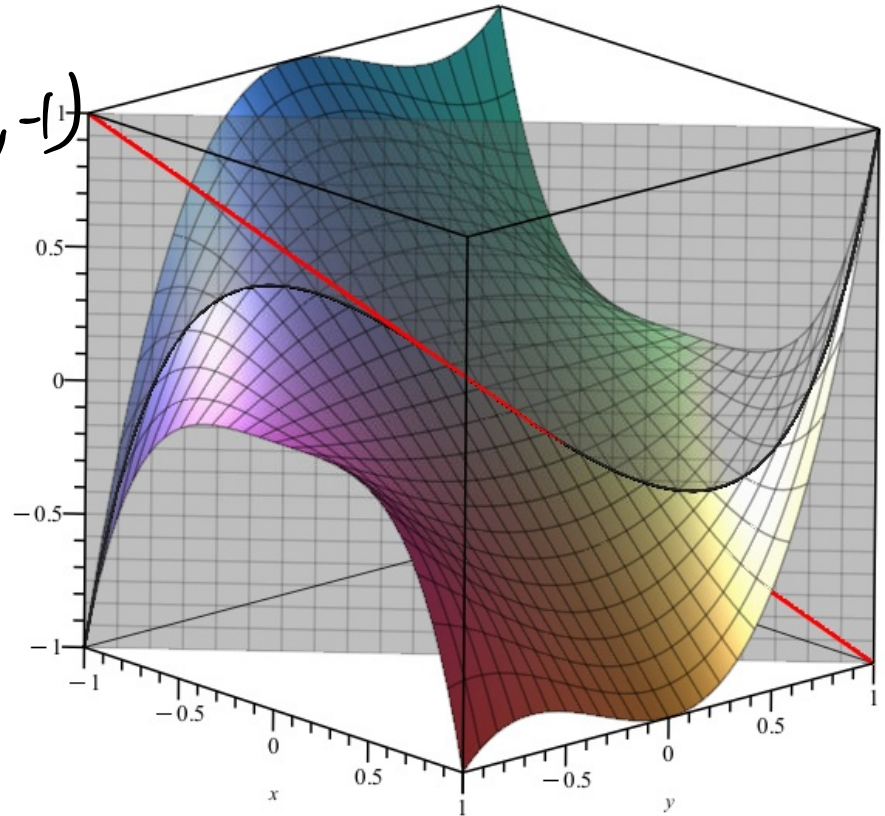


Figure: The tangent line to the surface at the origin in the direction of $(1,1)$ shown in red.

An Interesting Example

Exercise. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Show that f is not continuous but $D_{\mathbf{v}}(0, 0)$ exists for all $\mathbf{v} \neq \mathbf{0}$. Additionally, show that there is no plane that contains all lines that are tangent to the graph of f at the origin.

We showed in chapter 5 that f is not continuous, and so cannot be differentiable. Suppose $\mathbf{u} = (a, b) \in \mathbb{R}^2$ satisfy $a^2 + b^2 = 1$. Then

$$\begin{aligned} D_{\mathbf{u}}(0, 0) &= \lim_{h \rightarrow 0} \frac{f(h\mathbf{u}) - f(0, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(ha)(hb)^2 - 0}{(ha)^2 + (hb)^4} \\ &= \lim_{h \rightarrow 0} \frac{ab^2}{a^2 + h^2b^4} = \begin{cases} 0 & \text{if } a = 0 \\ \frac{b^2}{a} & \text{if } a \neq 0 \end{cases} \end{aligned}$$

An Interesting Example

Ex. cont.

Look at $\underline{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\underline{v}_2 = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}$, $\underline{v}_3 = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix}$

$$D_{\underline{v}_1} f(0,0), D_{\underline{v}_2} f(0,0) = \frac{(3/5)^2}{4/5} = \frac{9}{20}$$

$$D_{\underline{v}_3} f(0,0) = -\frac{9}{20}$$

So vectors that are parallel to the respected tangent lines are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4/5 \\ 3/5 \\ 9/20 \end{pmatrix}, \begin{pmatrix} -4/5 \\ 3/5 \\ -9/20 \end{pmatrix}$$

which are linearly independent, so there is no plane that contains all tangent lines at the origin.

